

Effect of increased sediment sulfide concentrations on the composition of stable sulfur isotopes ($\delta^{34}\text{S}$) and sulfur accumulation in the seagrasses *Zostera marina* and *Posidonia oceanica*

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Abstract

The effect of increased sediment sulfide concentrations on the sulfur isotopic composition ($\delta^{34}\text{S}$), total sulfur (TS) and elemental sulfur (S^0) concentrations in plant tissues was studied for the two seagrasses *Zostera marina* (3 weeks in laboratory) and *Posidonia oceanica* (4 months *in situ*). Porewater sulfide concentrations were experimentally regulated and plants exposed to high sediment sulfide concentrations had $\delta^{34}\text{S}$ signals closer to the $\delta^{34}\text{S}$ of sulfide, whereas plants exposed to no / low sulfide concentrations had $\delta^{34}\text{S}$ signals closer to the $\delta^{34}\text{S}$ of seawater sulfate, indicating a higher sulfide invasion in plants exposed to high sulfide concentrations. The $\delta^{34}\text{S}$ varied between the plant tissues in both species with the leaves having more positive $\delta^{34}\text{S}$ signals than roots and rhizomes, indicating that sulfide was invading into the roots and moved to the other tissues through the lacunae. TS and S^0 concentrations were higher in plants exposed to sulfide in both experiments suggesting that sulfur derived from sediment sulfide accumulated in the plants. The $\delta^{34}\text{S}$ signal in S^0 was similar to sediment sulfide verifying that S^0 found in the seagrasses originated from sediment sulfide. Direct comparisons of $\delta^{34}\text{S}$ in the two different seagrasses and across the treatments were not possible due to large differences in $\delta^{34}\text{S}$ of the sulfur sources. F_{sulfide} adjusted for these differences and may be a useful alternative, when $\delta^{34}\text{S}$ of the sulfur sources varies between study sites. There were no significant effects of sulfide exposure on plant growth and mortality in *Z. marina* and *P. oceanica* after 3 weeks and 8 weeks exposure, respectively, but *P. oceanica* showed indications of reduced growth and higher mortality after 16 weeks of sulfide exposure probably due to sulfide invasion/toxicity.

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1. Introduction

Sulfide is a potent phytotoxin affecting electron transport in respiration and photosynthesis and various other processes (Bagarinao, 1992). The combined effects of sulfide and plant anoxia have been suggested as key factors behind seagrass die offs events (Carlson et al., 1994), but current knowledge on sulfide toxicity in seagrasses is limited and tolerance seems to vary between species (Terrados et al., 1999; Erskine and Koch, 2000).

Seagrasses are adapted to anoxic and highly reduced sediments by having a specialized system of air channels (lacunae), which allow for diffusion of oxygen from the leaves to the rhizomes and roots (Roberts et al., 1984; Greve et al., 2003). Despite being buried in anoxic sediment, these tissues therefore have aerobic metabolism, which is energetically more favorable than anaerobic metabolism and does not produce toxic end products (Smith et al., 1988). Oxygen may leak from the roots into the sediment to create an oxic sphere, mainly around the root tip, where sulfide and other reduced phytotoxins are oxidized and thereby prevented from doing plant damage (Connell et al., 1999; Jensen et al., 2005; Frederiksen and Glud, 2006). The oxygen transport from the

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leaves may, however, under some circumstances (low oxygen concentrations in the water column and absence of photosynthesis) be insufficient to maintain an oxic rhizosphere and sulfide invasion into the plants occurs (Pedersen et al., 2004; Frederiksen and Glud, 2006). The sulfide moves rapidly between the plant compartments by gas phase diffusion in the lacunae (Pedersen et al., 2004; Borum et al., 2005) but very little is known about the fate of sulfide after it enters the plant. Holmer et al. (2005) found increased S^0 concentrations in roots and rhizomes of *Z. marina* exposed to high sulfide concentrations suggesting oxidation of sulfide to S^0 to be an important process. Other possible re-oxidation products include sulfate (SO_4^{2-}), which eventually may be incorporated into organic S-containing compounds such as glutathione or cysteine (Rennenberg, 1984; Raven and Scrimgeour, 1997; Wang et al., 2006). Yet another way for the seagrasses to remove sulfide is by passive diffusion of sulfide to the water column via the lacunae in the plant (Rennenberg, 1984) but to our knowledge, the importance of this process is not known for seagrasses.

Stable sulfur isotopes have been used to study sulfide uptake and transport in aquatic macrophytes (Carlson and Forrest, 1982; Fry et al., 1982; Raven and Scrimgeour, 1997; Frederiksen et al., 2006). The main sulfur sources for seagrasses are seawater sulfate, porewater sulfate and porewater sulfide and they all have different isotopic composition. When sulfate is reduced to sulfide during bacterial sulfate reduction, the ratio of the two stable sulfur isotopes ^{32}S and ^{34}S (expressed as $\delta^{34}S$) is changed due to bacterial fractionation and the produced sulfide ends up with a $\delta^{34}S$ signal more negative than sulfate. Seagrasses that have taken up sulfide therefore should have a different $\delta^{34}S$ signal than seagrasses not exposed to sulfide. The $\delta^{34}S$ of seawater sulfate is nearly constant around +21‰ in the world's oceans (Rees et al., 1978). In the sediment, the processes affecting $\delta^{34}S$ of the sulfur sources are many and complex (Canfield, 2001) and large variation exists. Sulfate in the porewater may potentially have highly positive $\delta^{34}S$ up to +60‰ (Kaplan et al., 1963; Jørgensen, 1979) but in natural sandy sediments is often just slightly more positive than $\delta^{34}S$ of seawater sulfate due to exchange of sulfate across the sediment-water interface (Canfield, 2001; Böttcher et al., 2004). The $\delta^{34}S$ of sulfides is also affected by the many processes involved in sediment sulfur cycling but values typically range from −10‰ to −20‰ with individual values as low as −50‰ (Strauss, 1997).

In this study, the effect of increased sediment sulfide concentrations on the composition of stable sulfur isotopes ($\delta^{34}S$), TS and S^0 concentrations in leaves, rhizomes and roots of the seagrasses *Zostera marina* and *Posidonia oceanica* was experimentally tested. *Z. marina*, is a fast growing species (Olesen and Sand-Jensen, 1994) capable of colonizing reduced sediments with high sulfide concentrations (Terrados et al., 1999). *P. oceanica* in contrast, is a slow growing species with long life span of up to 30 years (Marba et al., 1996), and is adapted to oligotrophic environments with low concentrations of sulfide. We therefore expected that susceptibility and tolerance to sulfide invasion varied between the species. The sediment conditions were manipulated for the 2 seagrass species by adding a carbon source to the sediments to enhance sulfide production (Terrados et al., 1999;

Holmer et al., 2005). *Z. marina* was transplanted into mesocosms and moved to the laboratory, where it was possible to control growth parameters such as light and temperature. Due to the sensitivity of *P. oceanica* to collection and laboratory conditions (Pérez, pers. com.) this experiment was done *in situ* and for practical reasons the number of treatments and sampled parameters were reduced compared to the experiment with *Z. marina*. Experiments with *Z. marina* lasted for 3 weeks, and therefore only relates to effects of sulfide over shorter timescales, whereas *P. oceanica* experiments ran for a period of 4 months. The $\delta^{34}S$ of the seagrasses was related to changes in TS and S^0 concentrations in the tissues to explore possible pathways of sulfide incorporation and oxidation after entering the plant. Finally, the effect of sulfide on growth and survival of the two species was investigated.

2. Materials and methods

2.1. Experiment with *Zostera marina* (Z.M.)

Plants of *Zostera marina* were collected in January 2004 from sandy sediment at the western part of Fyn, Denmark, at approximately 1 m water depth. The plants were gently washed free of sediment and transported to the laboratory together with sediment that had been sieved through a 1 mm sieve. Shoots were planted at a density of 870 shoots m^{-2} in 9 circular aquaria (i.d. 21 cm) after removal of dead leaves and rhizome tissue older than 5 internodes. In three of the aquaria (high sulfide, HS), caramelized glucose (580 g m^{-2}) was added to the sediment at approximately 5 cm depth to stimulate bacterial sulfate reduction and hence, increase the sulfide production (Terrados et al., 1999). In one of the glucose amended aquaria the porewater sulfide concentration increased much more than in the two others and results from this aquarium are shown separately (very high sulfide, HS+). Another three aquaria (OX) continuously pumped oxic seawater (air saturation) through the sediment by a peristaltic pump (24 ml/min) to create oxic sediment conditions with very low porewater sulfide concentrations (<1.5 μM). The last three aquaria (low sulfide, LS) were left untreated and porewater sulfide concentrations in between the two other treatments developed. The aquaria were placed in a large tank (500 l) with seawater from the Great Belt having approximately the same salinity (20‰) as the sampling site and about 1/3 of the water was changed every week. The light intensity was $\sim 200 \mu mol \text{ photons } m^{-2} s^{-1}$ in a 12:12 h diurnal cycle and the water temperature was 12–15 °C. The experiment ran for three weeks after which the plants were harvested and separated into leaves (the two youngest), rhizomes and roots, rinsed in deionized water and freeze dried. Plant growth was determined by the leaf puncturing technique ($n=10$, Zieman, 1974) before harvesting and sediment samples for determination of porewater sulfide concentrations, sulfur pool sizes and $\delta^{34}S$ were collected (see below).

2.2. Experiment with *Posidonia oceanica* (P.O.)

The experiment on *Posidonia oceanica* was done *in situ* at 10 m depth in Medes Islands, situated about 1 mile off the North East coast of Spain. The *P. oceanica* meadow had an average shoot density of 348 shoots m^{-2} (SD ± 71) and the sediment

Table 1
Concentrations of dissolved porewater sulfide (PW) and particulate AVS, and CRS pools±SE in sediments with *Z. marina* from the different experimental treatments (see text)

Sulfide pool	HS+	HS	LS	OX
PW sulfide (μM)	1745.2	331.3±8.5	125.4±52.9	b.d.
AVS (μmol S cm ⁻³)	0.4	0.5±0.1	0.6±0.2	b.d.
CRS (μmol S cm ⁻³)	5.8	6.5±0.7	8.2±0.6	4.6±0.3

In the OX treatment PW sulfide and AVS sulfide pool were below the detection limit (<0.1 μmol S cm⁻³, b.d.). Number of replicates were n=3 for OX and LS, n=2 for HS and n=1 for HS+.

was characterized by a high carbonate content of about 82% (Invers et al., 1995), a low organic matter content ranging from 1.03–1.19% DW (Invers et al., 1995) and positive redox potential (Alcoverro et al., 1995). For practical reasons only two treatments, high sulfide (HS) and low sulfide (LS), were compared. Three randomly selected 50x50 cm plots (HS) were installed and caramelized sucrose added to the sediment once a week (260 g plot⁻¹) from March to May and every two weeks from May to July to develop and maintain high sulfide concentrations. Another three plots (LS) received no treatment and acted as control. In addition to these treatments 3 plots were installed where sucrose + a slow releasing commercial fertilizer was added biweekly (40 g N and 40 g P per plot) for a study of the combined effects of nutrient enrichment and sulfide exposure on *P. oceanica* nutrient status (see Pérez et al., 2007). Plants from these plots were included with HS data to increase the number of replicates. Results where full replication (n=3) was available for both treatments suggests no additional effect of the nutrient addition on plant sulfur parameters (no significant differences between the two treatments in δ³⁴S and TS of leaves, S⁰ for leaves, root and rhizomes and sediment AVS and CRS pools, data not shown). Results of plant growth and mortality are based only on data from HS plots as the nutrient addition affected this parameter (Pérez et al., 2007).

The experiment started in March 2002 and ran for 4 months with samplings after 8 weeks in May and 16 weeks in July. At each sampling, plant growth was measured by leaf puncturing (n=10), according to Zieman (1974) modified by Romero (1989), and mortality was determined by counting the number of shoots in each plot and relating it to the initial number of shoots at the onset of the experiment. Plants for laboratory analyses were removed from the sediment, rinsed in deionized water and divided into leaves (the two youngest), rhizomes and roots, and then freeze dried. Sediment samples were collected and stored for later analysis (see below).

Table 2
AVS and CRS sulfur pools in May and July±SE in sediment (0–5 and 5–10 cm) with *P. oceanica* exposed to high (HS) and low (LS) sulfide levels

Sulfur pool	Depth (cm)	May		July	
		HS	LS	HS	LS
AVS (μmol S cm ⁻³)	0–5	20.2±1.3 ^a	0.4±0.1 ^b	26.8±4.5 ^a	0.3±0.1 ^b
	5–10	20.7±1.3 ^a	2.3±1.3 ^b	28.7±2.1 ^a	1.7±1.4 ^b
CRS (μmol S cm ⁻³)	0–5	21.4±4.4 ^a	12.2±3.1 ^a	39.8±17.1 ^a	8.8±2.0 ^a
	5–10	28.4±6.8 ^a	17.4±16.4 ^a	44.1±8.1 ^a	29.1±15.6 ^a

Number of replicates were n=3 for LS and n=6 for HS treatments. Letters indicate significant differences (p<0.05) between LS and HS treatments in May and July, respectively.

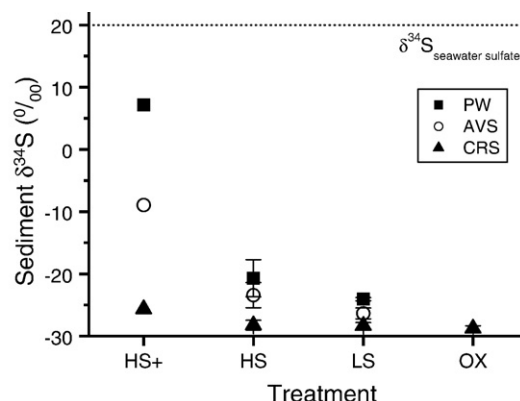


Fig. 1. Isotopic composition of stable sulfur isotopes (δ³⁴S) in porewater (PW), AVS and CRS sulfur pools of sediments with *Zostera marina* with different levels of sulfide concentrations (see text for further explanation). Error bars indicate SE (n=3 for OX and LS, n=2 for HS, n=1 for HS+). Dotted line indicates δ³⁴S for seawater sulfate.

2.3. Sediment analyses

Porewater for the analysis of sulfide concentrations and δ³⁴S of sulfide (only measured in the experiment with *Z. marina*) was extracted using a closed steel syringe with 0.4 mm holes drilled near the end (see Berg and McGlathery, 2001). To determine the sulfide concentration, 3 ml of porewater was extracted, preserved with zinc acetate and analyzed spectrophotometrically according to the method of Cline (1969). In both experiments, the presence of sulfide was monitored every week by inserting silver sticks into the sediment from each treatment (n=3) and sulfide was detected by the precipitation of black Ag₂S. For the δ³⁴S of sulfide, 30–40 ml of porewater were extracted and preserved with zinc acetate as ZnS. The ZnS was weighed into tin capsules together with vanadium pentoxide and analyzed by elemental analyzer combustion continuous flow isotope ratio mass spectroscopy (EA-C-CF-IRMS) at Iso-analytical, United Kingdom. The stable isotopic signatures were reported in standard delta notation (units per mill, ‰) as:

$$\delta^{34}\text{S} = \left[\left(\frac{R_{\text{sample}}}{R_{\text{standard}}} \right) - 1 \right] \times 1000$$

where R=³⁴S/³²S. The international standard for δ³⁴S is the Canyon Diablo Troilite (CDT) a meteorite of FeS used as a standard zero point for expression of sulfur isotopes.

Cores of sediment (n=3, i.d. 1.5 cm and 4 cm at *Z. marina* and *P. oceanica*, respectively) were taken before the plants were harvested and preserved with zinc acetate. In the *Z. marina* experiment, sediment from the root zone (1–7 cm) was used

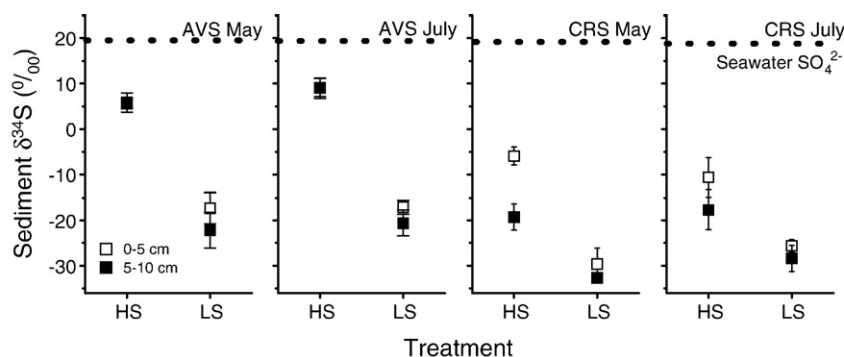


Fig. 2. Sulfur isotopic composition ($\delta^{34}\text{S}$) of AVS and CRS sulfur pools in May and July in sediments (0–5 and 5–10 cm) with *Posidonia oceanica* exposed to high (HS) and low (LS) sulfide levels. The $\delta^{34}\text{S}$ of seawater sulfate is indicated by the dotted line. Errors bars indicate standard errors ($n=3$ for LS and $n=6$ for HS).

whereas in the *P. oceanica* experiment, sediment was split up in 0–5 cm and 5–10 cm intervals. The sediment was distilled according to the two-step procedure of Fossing and Jørgensen (1989) where acid volatile sulfur (AVS), consisting of FeS and free sulfide, is extracted in the first step and chromium reducible sulfur (CRS), consisting of FeS_2 and S^0 , in the second step. The extracted sulfide was precipitated as Ag_2S and analyzed for $\delta^{34}\text{S}$ as described above. The size of the two sulfur pools was determined by weighing the precipitates.

Samples for measuring the $\delta^{34}\text{S}$ of sulfate in seawater and sediment porewater were prepared by centrifuging seawater or sediment (~ 3000 g, 10 min) followed by boiling of the supernatant under acidic conditions (3 M HCl) and precipitating sulfate with BaCl_2 as BaSO_4 .

2.4. Plant analyses

For the sulfur isotopic ($\delta^{34}\text{S}$) analysis, fine grained leaf, rhizome and root material was weighed into tin capsules together with vanadium pentoxide and analyzed for $\delta^{34}\text{S}$ by EA-C-CF-IRMS as described above. This analysis also measured the total concentration of sulfur (TS) in the plant tissue. The concentration of elemental sulfur (S^0) was determined by HPLC according to Zopfi et al. (2001) on dried plant tissue (5–20 mg) extracted in 5 ml HPLC-grade methanol for 24 hours.

The $\delta^{34}\text{S}$ of S^0 in the roots and rhizomes (only *Z. marina*) was determined by a modified soxhlet extraction of S^0 followed by isotopic analysis (Berner, 1964; Canfield et al., 1998). Plant tissue (350–650 mg) was placed in methanol for 24 hours and then centrifuged (~ 3000 g, 10 min). The supernatant was transferred to a solution with 8 ml 6 N HCl, 16 ml 1 M Cr^{2+} and Cu grains

(Merck 230 mesh ASTM) and distilled by boiling for 1 hour. During distillation, S^0 was reduced to H_2S which was precipitated with AgNO_3 as Ag_2S . The Ag_2S was analyzed for $\delta^{34}\text{S}$ as described for the sediment material. As considerable amounts of plant tissue were needed for this analysis, the 3 replicates from each treatment were pooled.

For both Z.M. and P.O. experiments the fraction of sulfur in the plants that originated from sediment sulfide (F_{sulfide}) was calculated as:

$$F_{\text{sulfide}} = \frac{\delta^{34}\text{S}_{\text{tissue}} - \delta^{34}\text{S}_{\text{sulfate}}}{\delta^{34}\text{S}_{\text{sulfide}} - \delta^{34}\text{S}_{\text{sulfate}}} \cdot 100$$

where $\delta^{34}\text{S}_{\text{tissue}}$ is the value measured in the leaf, rhizome or root and $\delta^{34}\text{S}_{\text{sulfate}}$ was taken as the values measured in the seawater. The $\delta^{34}\text{S}_{\text{sulfide}}$ was taken as the values measured in the AVS pools as this pool, with its relatively high reactivity and turnover in between porewater and CRS pools (Fossing and Jørgensen, 1989), was assumed to be the best representative of sulfide $\delta^{34}\text{S}$ over the entire course of the experiment. In the OX treatments of the Z.M. experiment, data from AVS pools were not available due to low pools and data from the LS treatment were used. For the P.O. experiment an average of $\delta^{34}\text{S}_{\text{AVS}}$ in the two depth intervals represented the $\delta^{34}\text{S}_{\text{sulfide}}$.

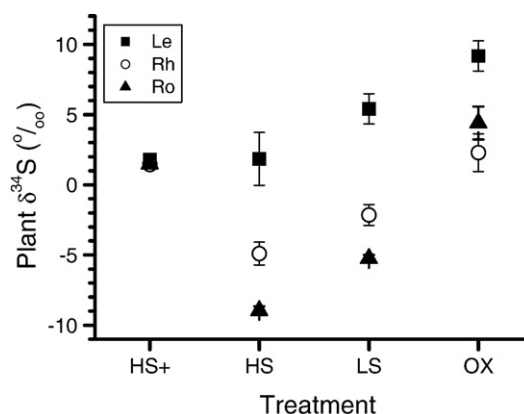


Fig. 3. The $\delta^{34}\text{S}$ of *Zostera marina* leaves (Le), rhizomes (Rh) and roots (Ro) exposed to different levels of sulfide in the sediment (HS+, HS, LS, OX, see text for further explanation). Error bars indicate SE ($n=3$ for OX and LS, $n=2$ for HS, $n=1$ for HS+).

Table 3

The $\delta^{34}\text{S}$ of sulfate in sediment porewater (0–10 cm) $\pm$ SE in May and July in HS and LS plots from the experiment with *P. oceanica*

Depth	$\Delta^{34}\text{S}$ May (‰)		$\delta^{34}\text{S}$ July (‰)	
	HS	LS	HS	LS
0–5 cm	+23.4 $\pm$ 0.9 ^a	+20.0 $\pm$ 0.2 ^b	+21.3 $\pm$ 0.5 ^a	+20.8 $\pm$ 0.1 ^a
5–10 cm	+23.7 $\pm$ 0.5	n.a.	+19.2 $\pm$ 2.5 ^a	+18.9 $\pm$ 0.8 ^a

Number of replicates are $n=3$ for LS and $n=6$ for HS. Letters indicate significant differences ($p < 0.05$) between LS and HS treatments in May and July, respectively.

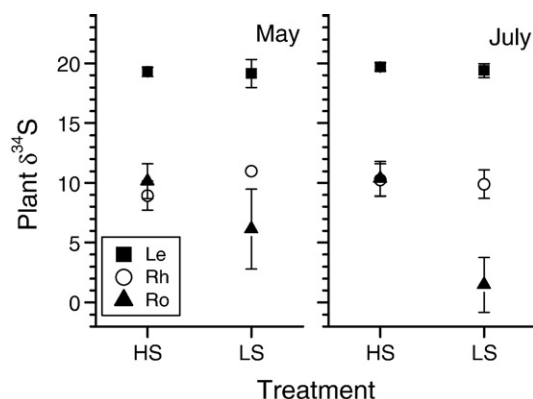


Fig. 4. The sulfur isotopic composition ($\delta^{34}\text{S}$) of leaves, rhizomes and roots of *Posidonia oceanica* exposed to high and low sulfide concentrations in the sediment. The two panels show $\delta^{34}\text{S}$ in May and July, respectively. Error bars indicate standard error ($n=2-3$ in May and 4 (roots+rhizome) and 6 (leaves) in July).

2.5. Statistical procedures

Data were tested for normality and then analyzed statistically using SAS ver. 8.02 (SAS Institute Inc.). A significance level of 95% ($p < 0.05$) was used.

In the Z.M. experiment the number of replicates was 1 (HS+), 2 (HS) and 3 (LS and OX) and it was therefore not possible to test for differences between all 4 treatments. HS+ was therefore excluded from statistical tests and remaining data were analyzed by 1-way ANOVA (AVS, CRS, plant growth, factor: treatment) or 2-way ANOVA (plant TS, S^0 , $\delta^{34}\text{S}$, F_{sulfide} , factor: treatment x plant compartment) followed by a Bonferroni adjusted Fisher's LSD test. For porewater concentrations of sulfide HS+ and HS treatments were pooled and analyzed by a non-parametric (Kruskal-Wallis) ANOVA test.

For the P.O. experiment data were analyzed statistically by a two sample independent t-test (AVS, CRS, $\delta^{34}\text{S}$ PW sulfate, TS, S^0 , plant growth, plant survival) and 2-way ANOVA ($\delta^{34}\text{S}$ and F_{sulfide} of plants, factor: treatment x plant compartment) and 3-way ANOVA ($\delta^{34}\text{S}$ of AVS and CRS factor: treatment x

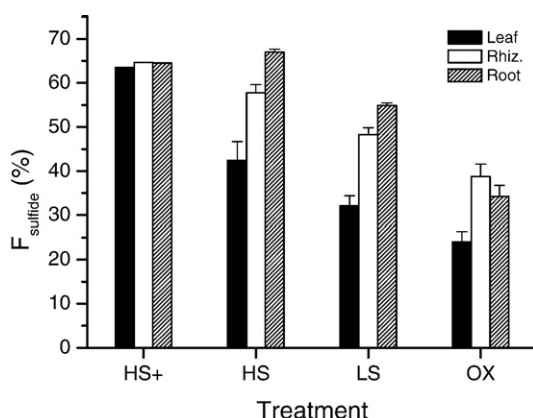


Fig. 5. The fraction of sulfur in leaves, rhizomes and roots of *Zostera marina* derived from sulfide in the sediment (F_{sulfide}). Error bars indicate SE ($n=3$ for OX and LS, $n=2$ for HS, $n=1$ for HS+).

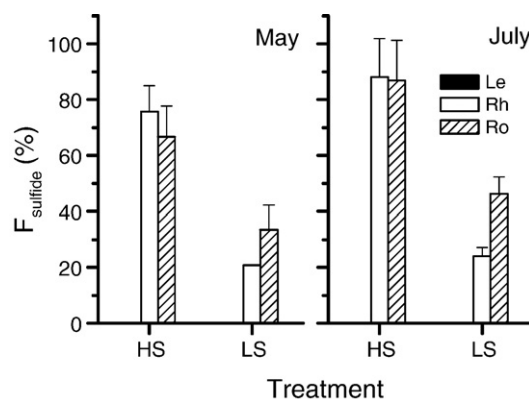


Fig. 6. The fraction of sulfur in leaves, rhizomes and roots of *Posidonia oceanica* derived from sulfide in the sediment (F_{sulfide}). Error bars indicate SE (May: $n=2-3$, July: $n=4$ (roots+rhizome) and 6 (leaves)).

pool x sediment depth) followed by a Bonferroni adjusted Fisher's LSD test.

3. Results

The different treatments affected the sulfide concentrations in the sediment substantially in both experiments. In the Z.M. experiment sulfide concentrations were around $330 \mu\text{M}$ in two of the glucose amended aquaria (HS) but in the third aquarium (HS+) sulfide concentrations became much higher ($\sim 1750 \mu\text{M}$) with high Ag_2S precipitation on the silver sticks (Table 1) In the untreated aquaria (LS) sulfide concentrations were lower ($\sim 125 \mu\text{M}$) while aquaria where the sediment was flushed with seawater (OX) had sulfide concentrations below the detection limit of the method of Cline (1969, $< 1.5 \mu\text{M}$) and no precipitation was found on silver sticks. Statistical analysis showed that sulfide concentrations were significantly different between the treatments (pooling the HS and HS+ treatments and using H_2S concentrations of $1.5 \mu\text{M}$ to represent OX treatments).

In the P.O. experiment sediment sulfide concentrations was monitored with silver sticks which were colored completely black from the sediment surface down to 10 cm depth in the HS treatment, indicating high sulfide concentrations in the porewater, whereas no precipitation of sulfide was observed in LS plots (data not shown).

Table 4
Total sulfur (TS) and elemental sulfur (S^0) concentrations $\pm$ SE in *Z. marina* leaves rhizomes and roots from the different treatments

Parameter	Tissue	HS+	Treatments		OX
			HS	LS	
TS ($\mu\text{mol gDW}^{-1}$)	Leaf	309.0	281.8 ± 7.3	234.0 ± 15.5	174.5 ± 7.1
	Rhiz.	509.4	424.5 ± 49.6	346.9 ± 29.7	195.4 ± 8.1
	Root	298.6	328.9 ± 44.5	281.6 ± 17.5	174.2 ± 17.4
S^0 ($\mu\text{mol gDW}^{-1}$)	Leaf	b.d.	b.d.	b.d.	b.d.
	Rhiz.	95.1	17.0 ± 2.5	10.9 ± 1.8	0.2 ± 0.03
	Root	138.6	39.4 ± 11.0	29.9 ± 4.2	0.7 ± 0.1

The S^0 concentration was below the detection limit ($< 0.1 \mu\text{mol gDW}^{-1}$) in leaves (b.d.). The number of replicates was $n=3$ for OX and LS, $n=2$ for HS and $n=1$ for HS+.

Table 5

Concentrations of TS and $S^0 \pm SE$ in leaves, rhizomes and roots of *P. oceanica* exposed to high (HS) and low (LS) sulfide concentrations in the sediment

Parameter	Tissue	May		July	
		HS	LS	HS	LS
TS ($\mu\text{mol gDW}^{-1}$)	Leaf	161.2 $\pm$ 3.1 ^a	134.1 $\pm$ 9.2 ^a	197.9 $\pm$ 1.4 ^a	166.3 $\pm$ 3.5 ^b
	Rhiz.	538.6	249.5 $\pm$ 15.0	484.6 $\pm$ 27.4 ^a	262.6 $\pm$ 74.3 ^a
	Root	296.3 $\pm$ 25.0 ^a	193.4 $\pm$ 24.9 ^a	385.5 $\pm$ 22.5 ^a	220.4 $\pm$ 35.6 ^b
S^0 ($\mu\text{mol gDW}^{-1}$)	Leaf	0.2 $\pm$ 0.02 ^a	0.5 $\pm$ 0.2 ^a	0.4 $\pm$ 0.1 ^a	0.2 $\pm$ 0.04 ^a
	Rhiz.	3.1 $\pm$ 0.6 ^a	3.1 $\pm$ 1.2 ^a	3.9 $\pm$ 0.9 ^a	3.2 $\pm$ 1.0 ^a
	Root	8.5 $\pm$ 1.2 ^a	1.4 $\pm$ 0.3 ^b	11.5 $\pm$ 1.3 ^a	2.0 $\pm$ 0.4 ^b

Number of replicates was $n=2-3$ in LS treatments. In HS treatments $n=4$ (roots + rhizome) and 6 (leaves), except for rhizome TS in May where $n=1$. Letters indicate significant differences ($p<0.05$) between LS and HS treatments in May and July, respectively.

The AVS sulfur pools in the sediment were similar in HS+, HS and LS treatments of the Z.M. experiment (around $0.5 \mu\text{mol S cm}^{-3}$, Table 1) and below the detection limit ($<0.1 \mu\text{mol S cm}^{-3}$) in the OX treatment. The CRS sulfur pools were generally about an order of magnitude larger than the AVS pools ranging from $4.6-8.2 \mu\text{mol S cm}^{-3}$ with no significant differences between the treatments. In the P.O. experiment the AVS sulfur pools were significantly ($p<0.05$) higher in HS treated plots in both May and July (range $20.2-28.7 \mu\text{mol S cm}^{-3}$) compared to LS plots (range $0.3-2.3 \mu\text{mol S cm}^{-3}$, Table 2). The CRS pools were always higher in HS compared to LS treatments but the difference was less pronounced and not significant with values ranging from $8.8-29.1 \mu\text{mol S cm}^{-3}$ in the LS plots to $21.4-44.1 \mu\text{mol S cm}^{-3}$ in the HS plots. There was no significant change with depth in the sediment in AVS and CRS.

The $\delta^{34}\text{S}$ of the PW sulfide and the AVS pool in the Z.M. experiment increased slightly, but not significantly, from the LS to the HS treatment (range -24.1% to -20.7% and -26.3 to -23.4% , respectively) whereas $\delta^{34}\text{S}$ of the HS+ was considerably higher in AVS sulfur (-8.9%) and PW sulfide ($+7.2\%$, Fig. 1). Due to low concentrations of sulfide, it was not possible to obtain enough material for $\delta^{34}\text{S}$ analysis in the OX treatment. The $\delta^{34}\text{S}$ of the CRS pool was more or less constant in all treatments ranging from -25.7 to -28.8% .

In the P.O. experiment the $\delta^{34}\text{S}$ of the AVS pool increased markedly from the LS plots (range -17.1% to -22.2%) to the HS plots (range $+5.6\%$ to $+9.1\%$) and the same pattern, however, less pronounced, appeared in the CRS pool with $\delta^{34}\text{S}$ ranging from -25.6% to -32.7% in LS plots and from -19.3% to -5.9% in HS plots (Fig. 2). Hence $\delta^{34}\text{S}$ was always significantly lower ($p<0.05$) in CRS pools compared to AVS sulfur pools. The $\delta^{34}\text{S}$ signal generally became more negative deeper in the sediment in both AVS and CRS sulfur pools, but only significantly ($p<0.05$) in May.

The $\delta^{34}\text{S}$ of sulfate was $+20.2\%$ to 20.4% ($n=3$) in seawater of the Z.M. experiment. The $\delta^{34}\text{S}$ of the porewater sulfate was only determined for the HS+treatment and was $+22.6\%$ but the $\delta^{34}\text{S}$ of porewater sulfate in LS and HS treatments measured in a pilot study similar to the present study, ranged from $+21.1$ to $+21.5\%$, suggesting no significant difference between the treatments. In the P.O. experiment the $\delta^{34}\text{S}$ of seawater sulfate was $+18.9$ to 19.0% ($n=3$) and porewater sulfate $\delta^{34}\text{S}$ ranged

from $+18.9\%$ to $+20.8\%$ in the LS plots and from $+19.2\%$ to $+23.7\%$ in the HS plots in the (Table 3).

The $\delta^{34}\text{S}$ of *Zostera marina* leaves (range $+9.2\%$ to $+1.2\%$) was significantly higher ($p<0.05$) than in roots and rhizomes (range $+4.4\%$ to -9.0% , Fig. 3). Roots generally had lower $\delta^{34}\text{S}$ than rhizomes but the difference was not significant. The plants from the HS+treatment were exceptional as no difference was found between $\delta^{34}\text{S}$ in leaves, rhizomes and roots ($\delta^{34}\text{S} \sim +1.6\%$). The $\delta^{34}\text{S}$ decreased from the OX to LS to HS treatments in all tissues, significant ($p<0.05$) between OX and LS plants in roots and rhizomes but not in leaves. The P.O. experiment showed similar variation in $\delta^{34}\text{S}$ between tissues. In LS plots $\delta^{34}\text{S}$ was higher in leaves ($\sim +19\%$), than in rhizomes ($\sim +10\%$) which again were more positive than roots ($+1\%$ in May and $+6\%$ in July, $p<0.05$ for July, Fig. 4). In the HS plants, $\delta^{34}\text{S}$ was significantly ($p<0.05$) different between leaves ($\delta^{34}\text{S} \sim +19\%$) and rhizomes ($\delta^{34}\text{S} \sim +10\%$) and not between rhizomes and roots ($\delta^{34}\text{S} \sim +10\%$). Significant differences between treatments only existed in the roots where $\delta^{34}\text{S}$ of roots in the HS plots was significantly ($p<0.05$, July) higher than the LS plots.

The F_{sulfide} (fraction of sulfur in the plants that originated from sediment sulfide) showed that *Zostera marina* roots and rhizomes derived a significantly higher fraction ($p<0.05$) of the sulfur from sulfide in the sediment ($34-67\%$) compared to leaves ($24-42\%$, Fig. 5). Further, the F_{sulfide} varied significantly ($p<0.05$) between treatments with the highest values in plants from the HS+ and HS treatments (pooling data from HS and HS+) and the lowest in OX plants. *P. oceanica* also showed a significant difference between the tissues ($p<0.05$) with roots and rhizomes deriving $21-88\%$ and $34-87\%$ of their sulfur from sulfide, respectively, whereas leaves derived none of their sulfur from sulfide (Fig. 6). The F_{sulfide} varied between HS and LS treatments in both roots and rhizomes with higher values in plants from the HS treatment, however, only significant ($p<0.05$) for rhizomes in July.

The total sulfur (TS) concentration in both *Z. marina* and *P. oceanica* was generally highest in rhizomes, intermediate in roots and lowest in leaves. In the Z.M. experiment TS increased from the OX treatment to the HS+treatment in all tissues ($p<0.05$ for OX, LS, HS comparisons, Table 4). Similarly, the P.O. experiment showed that TS concentrations were markedly higher in roots, rhizomes and, to less extent, in leaves (significant in July, $p<0.05$) of the HS plants compared to LS plants

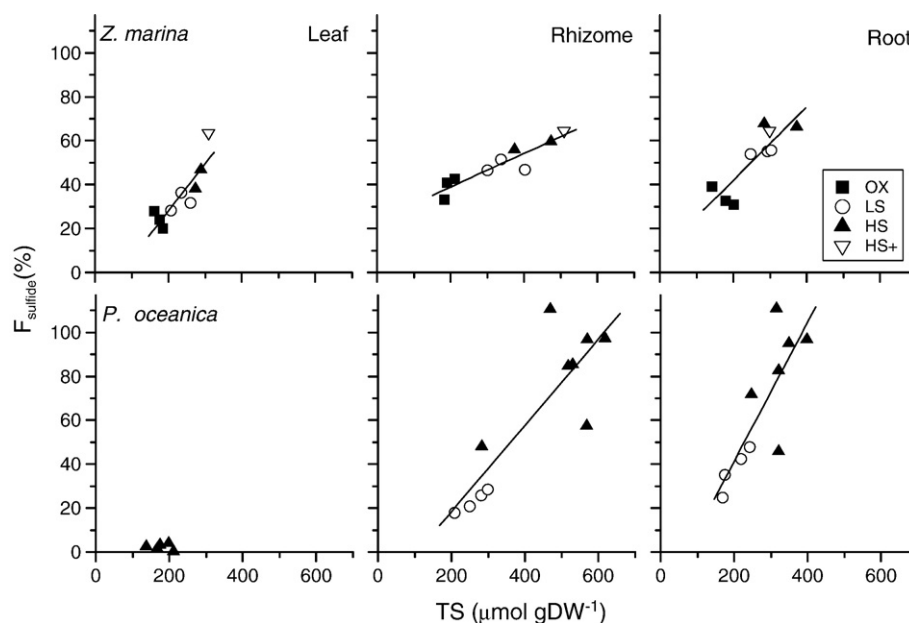


Fig. 7. F_{sulfide} as a function of TS concentration in leaves, rhizomes and roots of *Z. marina* and *P. oceanica*. For *Z. marina* linear regression analysis showed a significant relation ($p < 0.05$) for roots ($R^2 = 0.73$), rhizomes ($R^2 = 0.86$) and leaves ($R^2 = 0.74$). For *P. oceanica* significant relations were significant ($p < 0.05$) for rhizomes ($R^2 = 0.74$) and roots ($R^2 = 0.66$). No relation was found for leaves of *P. oceanica* as F_{sulfide} was always close to 0.

(Table 5). Linear regression analysis of F_{sulfide} as a function of TS concentration in *Z. marina* showed a significant ($p < 0.05$) positive relation in roots ($R^2 = 0.73$), rhizomes ($R^2 = 0.86$), and leaves ($R^2 = 0.74$), indicating that plants with a large part of the sulfur being derived from sulfide accumulated more sulfur (Fig. 7 upper panel). The same relationship appeared for *P. oceanica* with a significant ($p < 0.05$) positive relation in roots and rhizomes ($R^2 = 0.66$, rhizomes $R^2 = 0.74$, Fig. 7 lower panel) but not in leaves as F_{sulfide} here was always close to zero (data not shown).

Elemental sulfur (S^0) concentrations showed a pattern similar to TS concentrations in both experiments. In the *Z. M.* experiment the S^0 concentration was higher in roots compared to rhizomes ($p < 0.05$) and leaves where concentrations were below the detection limit (Table 4). The S^0 concentration increased significantly from the OX to the HS+ treatment in roots and rhizomes ($p < 0.05$ between OX and LS, and OX and HS treatments). The $\delta^{34}\text{S}$ of S^0 extracted from the plant tissue ranged from -14.2 to -16.7% and was always markedly lower than $\delta^{34}\text{S}$ of the TS in the plant (Table 6). The S^0 concentrations of *P. oceanica* were generally lowest in the leaves in both May and July with concentrations ranging from 0.2 – $0.5 \mu\text{mol S gDW}^{-1}$ (Table 5). Roots and rhizomes had similar S^0 concentrations (range 1.4 to $3.2 \mu\text{mol S gDW}^{-1}$) in plants from the LS plots, whereas in the HS plots roots had significantly ($p < 0.05$) higher S^0 concentrations (range 8.5 – $11.5 \mu\text{mol S gDW}^{-1}$) than rhizomes (range 3.1 – $3.9 \mu\text{mol S gDW}^{-1}$) in both May and July. The effect of the treatments was, however, only significant in roots, with the highest S^0 concentration in the HS plots ($p < 0.05$). Generally, the concentration of S^0 in *P. oceanica* was low constituting $< 3\%$ of the TS concentration.

The effect of increasing sulfide concentrations in the sediment on plant $\delta^{34}\text{S}$ and TS was tested for *Z. marina*. The $\delta^{34}\text{S}$ showed a

significant ($p < 0.05$) negative linear relationship with sediment sulfide concentrations in all tissues (leaves: $R^2 = 0.64$, rhizomes: $R^2 = 0.76$, roots: $R^2 = 0.73$, excluding the value from the HS+ treatment, Fig. 8, left panel). The TS concentration also showed a significant ($p < 0.05$) positive relationship with the sediment sulfide concentration in all tissues (leaves: $R^2 = 0.90$, rhizomes: $R^2 = 0.69$, roots: $R^2 = 0.67$, Fig. 8 right panel) for OX, LS and HS treatments indicating for both relationships that sulfur accumulation was higher in plants exposed to high concentrations of sediment sulfide. However, including data from the HS+ treatment in regressions showed that the relationship is not linear at high sediment sulfide concentrations.

The growth of *Z. marina* over three weeks was not significantly different among the treatments and ranged from $0.019 \text{ mm mm}^{-1} \text{ d}^{-1}$ in HS+ to $0.023 \text{ mm mm}^{-1} \text{ d}^{-1}$ in OX (Table 7), and no significant plant mortality was observed during the experiment.

The mortality of *P. oceanica* was noticeably higher and leaf growth rates were lower in plants from the HS plots compared to LS plots, but the differences were not significant due to high

Table 6
Comparison of $\delta^{34}\text{S}$ values of TS and elemental sulfur in rhizome and roots of *Z. marina* from LS and HS treatments

Treatment	Tissue	$\delta^{34}\text{S}_{\text{total sulfur}}$	$\delta^{34}\text{S}_{\text{elemental sulfur}}$
HS	Root	-9.0	-16.7
	Rhizome	-4.9	-15.7
LS	Root	-5.3	-14.2
	Rhizome	-2.2	-15.1

Samples from three replicates are pooled. No data available from OX and HS+ treatments due to low amounts of plant material.

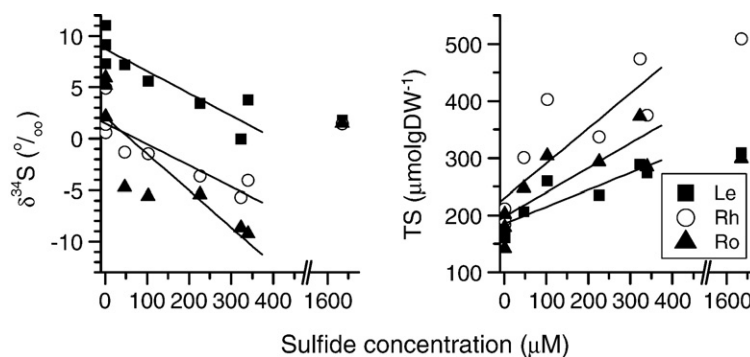


Fig. 8. The $\delta^{34}\text{S}$ (left panel) and TS concentrations (right panel) in leaves, rhizomes and roots of *Z. marina* as a function of sulfide concentration in the sediment. Linear regression analysis of $\delta^{34}\text{S}$ vs sulfide concentrations (excluding the values at 1745 μM) showed significant relationships ($p < 0.05$) for leaves ($R^2 = 0.64$), rhizomes ($R^2 = 0.76$) and roots ($R^2 = 0.73$). Regressions for TS vs sulfide concentrations (excluding the values at 1745 μM) were also significant ($p < 0.05$) for leaves ($R^2 = 0.90$), rhizomes ($R^2 = 0.69$) and roots ($R^2 = 0.67$).

variation among the replicates (Table 8). In the HS treatment plant mortality was 38% when the experiment finished in July, whereas it was only 9% in the LS treatment. Leaf growth rates were slightly reduced in HS plots in May, but not in July (Table 8).

4. Discussion

The different treatments significantly affected sediment conditions in both experiments with higher sulfide concentrations and sulfur pools in sediments where organic carbon was added. Especially the experiment with *P. oceanica* showed a substantial increase in the AVS sulfur pool between the treatments, and attained pools similar to high impacted sediments in fish farm surroundings, where high porewater sulfide pools have been measured (Frederiksen et al., 2007). The porewater sulfide concentration in the experiment with *Z. marina* increased from below detection to 125 μM and up to 1.8 mM, which are several folds higher than measured in natural sediments (Holmer and Nielsen, 1997).

The $\delta^{34}\text{S}$ of the seagrasses generally changed in response to the treatments. In the *Z. marina* experiment the $\delta^{34}\text{S}$ decreased from OX to the HS treatments in all tissues indicating that sulfide invasion in the plants increased when sulfide concentrations in the sediment were high. Accordingly, a significant inverse relation was found between $\delta^{34}\text{S}$ of the leaves and rhizomes and sulfide concentration in the sediment. Interestingly, the $\delta^{34}\text{S}$ of plants *Z. marina* from the HS+treatment and *P. oceanica* from the HS treatment showed no response or even increased with increased sulfide exposure, which under normal circumstances would indicate less sulfide invasion. However, in order to make direct comparisons of $\delta^{34}\text{S}$ signal in seagrasses from different treatments or different locations requires that the

$\delta^{34}\text{S}$ of the sulfur sources is the same among the treatments/locations. This was generally the case for sulfate in the seawater and porewater, but the $\delta^{34}\text{S}$ of sulfide increased considerably in the HS+treatment in the *Z. marina* experiment and in the HS treatments of the *P. oceanica* experiment. The increase in $\delta^{34}\text{S}$ of sulfide could result from a decreased fractionation of sulfate by sulfate reducing bacteria at high rates of sulfate reduction (Canfield, 2001), but the factors controlling isotopic fractionation of sulfide in marine sediments are many and complex (Canfield, 2001) and to fully explain the changes in $\delta^{34}\text{S}$ of sulfide require more detailed analyses.

The uptake of more positive sulfide resulted in relatively smaller changes in $\delta^{34}\text{S}$ of these plants, and direct comparisons between treatments are therefore not possible. The variation in $\delta^{34}\text{S}$ of the sulfur sources can, however, be adjusted for by calculating the F_{sulfide} , which determines the fraction of sulfur in the plants derived from sulfide. This parameter is based on the measured $\delta^{34}\text{S}$ value of the sulfide and sulfate sources, and the F_{sulfide} therefore allows direct comparisons between treatments and experiments. In contrast to the direct comparisons of $\delta^{34}\text{S}$, F_{sulfide} showed a clear effect of the treatments in both experiments. Hence, in the *Z. marina* experiment, the F_{sulfide} nearly doubled in all tissues from treatments with the low to the high sulfide exposure and the same applied to roots and rhizomes from the *P. oceanica* experiment. This suggests that significant sulfide invasion happened in both *Z. marina* and *P. oceanica* when sulfide exposure in the sediment increased. Similarly, Oakes and Connolly (2004) found a close correlation between the $\delta^{34}\text{S}$ of *Zostera capricornii* and sediment sulfide concentrations under experimental conditions, whereas Frederiksen et al. (2006) found this relationship to be less strong in

Table 7

Leaf growth of *Z. marina* exposed to different sulfide concentrations in the sediment during 4 weeks (see text for further explanation) $\pm$ SE ($n = 3$ for LS and OX, $n = 2$ for HS, $n = 1$ for HS+)

	HS+	HS	LS	OX
RGR ($\text{mm mm}^{-1} \text{d}^{-1}$)	0.019	0.021 ± 0.001	0.020 ± 0.0002	0.023 ± 0.002

Table 8

Leaf growth and shoot mortality of *P. oceanica* in LS and HS treatments $\pm$ SE

	HS		LS	
	May	July	May	July
RGR ($\text{mg DW shoot}^{-1} \text{d}^{-1}$)	4.1 ± 1.6	7.3 ± 1.2	6.5 ± 1.0	6.8 ± 1.0
Plant survival (%)	93.7 ± 10.8	62.4 ± 17.6	98.8 ± 2.9	91.4 ± 4.4

The number of replicates was $n = 3$ for both treatments. There were no significant differences between May and June in the treatments ($p > 0.05$).

natural beds of *Z. marina*. Although important, sulfide invasion in seagrasses is therefore not controlled exclusively by the sulfide concentration in the sediment, but is also affected by a range of environmental and plant related parameters such as light and oxygen status in the water column (Pedersen et al., 2004; Borum et al., 2005), and probably many other factors (temperature, water current regime, plant morphology).

The below-ground tissues of *Z. marina* and *P. oceanica* generally had $\delta^{34}\text{S}$ closer to the $\delta^{34}\text{S}$ of the sediment sulfide pools and higher F_{sulfide} than leaves suggesting that sulfide entered the plants via the roots and moved to the rhizomes and leaves. This is in accordance with microelectrode studies, which have shown the presence of gaseous sulfide in the lacunae of *Z. marina* and *Thalassia testudinum* (Pedersen et al., 2004; Borum et al., 2005). Interestingly, the $\delta^{34}\text{S}$ values and F_{sulfide} suggested, that leaves in *P. oceanica* did not derive any or very little of their sulfur from sulfide, even in the plots with high sulfide concentrations in the sediment. The same has been observed for *P. oceanica* in the vicinity of fish farms with high sulfide pools in the sediment (Frederiksen et al., 2007). *Z. marina* in contrast had $\delta^{34}\text{S}$ of leaves significantly below $\delta^{34}\text{S}$ values of seawater sulfate, and even in plants from the OX treatment, exposed to very low sulfide concentrations, 24% of the sulfur was derived from sulfide. Similar findings of low $\delta^{34}\text{S}$ in leaves have been reported in natural beds of *Z. marina* (Frederiksen et al., 2006), and sulfide or derivatives from sulfide therefore seems to move more freely in *Z. marina* compared to *P. oceanica*. The presence of gaseous sulfide in the leaves may have important implications for the plant because it implies that sulfide is passing the leaf meristem, which has low tolerance to sulfide and anoxic conditions (Greve et al., 2003). Morphological differences between the species could be responsible for these differences. *P. oceanica* has both horizontal and vertical rhizomes and the diffusion distance from the roots to the leaves is therefore longer in this species compared to *Z. marina*, which does not have vertical rhizomes. Also, the lacunal system for transporting gasses in the plants may be more well developed in *Z. marina*, which generally colonize highly reduced sediments and therefore needs an efficient transport of oxygen from the leaves to the below-ground tissues. *P. oceanica*, in contrast, grows in oligotrophic areas with less sulfide exposure from the sediments and less efficient transport of oxygen is probably sufficient under normal conditions. Microscopical inspection of transverse sections of rhizomes of both species indicated a larger number of lacunal cells in *Z. marina* (data not shown), but further studies with direct quantification of lacunal volume are needed to document this.

It could be argued that the changes in $\delta^{34}\text{S}$ were not caused by uptake and accumulation of sulfides inside the plant, but was an artifact from precipitation of sulfides and oxidized sulfur compounds on the outer surface of the plant (Fry et al., 1982). For *Z. marina*, however, there is strong evidence that sulfide did enter the plants. First of all gaseous sulfide has been measured directly inside the rhizomes of *Z. marina* with microelectrodes (Pedersen et al., 2004) and, furthermore, changes in $\delta^{34}\text{S}$ also occurred in the leaves, which were not in direct contact with sulfide on the outer surfaces. Sulfides precipitating on the outer surfaces of the root and rhizomes could, however, still have

influenced our results, but a washing experiment where the roots and rhizomes were washed in three different agents with increasing ability to dissolve the sulfide precipitates (reagent 1: distilled H_2O , reagent 2: 6 M HCl to remove FeS, reagent 3: 1 M Cr^{2+} to remove FeS_2 and S^0) and subsequently analyzed for sulfur content showed very little precipitation on the surfaces and the main part of precipitates were removed by washing with distilled water (Frederiksen et al., 2006). Furthermore, a precipitation of iron-sulfur compounds on the surface of *Z. marina* roots exposed to high sediment sulfide concentrations was investigated by Holmer et al. (2005) and they found no precipitation. Thus, it is unlikely that external precipitation of sulfides on roots and rhizomes caused the changes in $\delta^{34}\text{S}$ of *Z. marina*. For *P. oceanica*, evidence is less strong as internal sulfide and oxygen dynamics to our knowledge has not previously been studied for this species. The lifespan of *P. oceanica* is much longer than for *Z. marina*, which generally leaves more time for precipitates to develop on the plant surfaces. It is therefore possible that some precipitation takes place on this species under natural conditions and could also have occurred during the 4 month experiment. The tissues were, however, thoroughly rinsed and for the rhizomes the scales were peeled off leaving only a small fraction of the surface tissue after the cleaning procedure. Further, Pérez et al. (2007) found clear indications of anaerobic metabolism in the roots and rhizomes of the examined plants, indicating that *P. oceanica* at times were anoxic and hence open for sulfide invasion.

The uptake of sulfate from the porewater with low $\delta^{34}\text{S}$ resulting from reoxidation of sulfide in the rhizosphere may also change the $\delta^{34}\text{S}$ of the seagrasses (Trust and Fry, 1992), but in the present studies the $\delta^{34}\text{S}$ of porewater sulfate only deviated slightly from $\delta^{34}\text{S}$ of seawater sulfate indicating that the pool of porewater sulfate was constantly renewed by exchange across the sediment-water interface. The observed changes in $\delta^{34}\text{S}$ of *Z. marina* and *P. oceanica* are therefore considered to be due to sulfide entering and accumulating in the plants.

4.1. Accumulation of sulfur compounds in seagrasses

The total sulfur concentrations (TS) of leaves, rhizomes and roots of *Z. marina* and *P. oceanica* also changed significantly by the sulfide treatments. TS increased in plants exposed to high sulfide concentrations in the sediment, but linear regressions were less strong than found for $\delta^{34}\text{S}$ of plants and sediment sulfide concentrations indicating a more complex relationship between sulfide exposure and TS accumulation.

A strong inverse relation between $\delta^{34}\text{S}$ and TS concentrations in the plant tissue has been found in natural beds of *Z. marina* (Frederiksen et al., 2006) indicating that sulfide or derivatives from sulfide accumulates in the plants when exposed to sulfide. Using F_{sulfide} as a substitute for direct measurements of $\delta^{34}\text{S}$ likewise showed a significant positive relation with TS in both *Z. marina* and *P. oceanica* indicating that more of the plant sulfur was derived from sediment sulfide in plants with high TS concentrations. One possible transformation of the invading sulfide is reoxidation by exposure to oxygen in the plant lacunae. We found increased levels of S^0 in roots and rhizomes of

Z. marina and roots of *P. oceanica* exposed to sulfide, suggesting S^0 to be one of the reoxidation products. This is supported by the fact that $\delta^{34}S$ analysis of S^0 extracted from the roots and rhizomes was markedly more negative than $\delta^{34}S$ of total plant sulfur and closer to $\delta^{34}S$ of sulfide. The same has been found for *P. oceanica* exposed to high sediment sulfide in fish farms surroundings (Frederiksen et al., 2007). Similar findings of S^0 and TS co-accumulation have been reported from other studies of *Z. marina* (Holmer et al., 2005; Frederiksen et al., 2006) and *P. oceanica* (Frederiksen et al., 2007). The S^0 concentration, however, could not account for all of the observed increase in TS between treatments in either seagrass species as linear regression of S^0 as a function of TS in *Z. marina* and *P. oceanica* did not show a 1:1 relationship (data not shown). The analyses of S^0 were conducted about 16 months after the plants were sampled and some reoxidation may have occurred in this period reducing the concentration of S^0 . However, other reoxidation products may also have been produced in the plant, such as sulfate, which has been shown to be a storage compound for excess sulfur in higher land plants (Rennenberg, 1984; Schmidt and Jäger, 1992). Finally, incorporation of excess sulfide into organic sulfur compounds or excretion of invading sulfide via the lacunae system in leaves to the water column are also a possibility as observed for land plants exposed to excess sulfur (Rennenberg, 1984). The importance of these processes is not known for seagrasses and needs further investigation.

The TS pools were surprisingly similar in the two seagrasses, and also the accumulations in response to sulfide exposure were quite similar where pools increased with a factor of 2–3 in rhizomes and roots compared to 20–75% in the leaves. This happened despite the different experimental settings (degree and duration of sulfide exposure) for the two seagrasses, and future studies are needed to explore these observations. The rhizomes showed the highest storage of TS, which could be related to the general function of rhizomes as storage organs for amino acids, carbohydrates and starch (Burke et al., 1996; Pérez et al., 2007; Vichkovitten et al., 2007).

4.2. Effects of sulfide on plant growth and survival

The two species seemed to have different responses to increased sulfide concentrations in the sediment, but direct comparisons should be made with caution due to the different duration of the experiments. Leaf growth and plant mortality of *Z. marina* was not affected by the treatments during the 3 week study, which is consistent with a similar study of *Z. marina* after three weeks of glucose addition (Holmer et al., 2005). Terrados et al. (1999) likewise did not find increased mortality two months after in situ addition of sucrose to the sediment of a Danish *Z. marina* bed, but leaf growth was reduced suggesting that long term exposure negatively affect *Z. marina* growth. Short term exposure to high sulfide concentrations in combination with low light or low oxygen concentrations in the water column on the contrary causes a significant reduction in growth and survival of *Z. marina* (Holmer and Bondgaard, 2001; Holmer and Nielsen, 2007; Holmer et al., 2005). This suggests that *Z. marina* is well capable of surviving and growing despite sulfide invasion in sediments

with high sulfide concentrations, but additional stress either from low light availability or low oxygen concentrations in the water column lead to mortality.

P. oceanica, in contrast to *Z. marina*, showed considerable mortality in plants from the HS plots, where 38% of the initial shoots had died after the 4 month study period. Leaf growth was, however, only reduced in HS plants in May, and increased again in July, indicating that the surviving plants had adapted to the higher sulfide concentrations. The longer period of exposure to sulfide in the *P. oceanica* study may have contributed to the increased mortality relative to the *Z. marina* study, but the changes in growth already appeared after 8 weeks in May suggesting that plants were affected early in the study. Increased mortality of *P. oceanica* has also been found in organic enriched sediments with high sulfide concentrations near marine fish farm facilities (Delgado et al., 1997, 1999; Pérgent et al., 1999; Diaz-Almela et al., in press) and in sediments with sulfide accumulating in the porewaters (Calleja et al., 2007) and mortality has been found to be significantly related to sulfide invasion in this species (Frederiksen et al., 2007).

5. Conclusion

In conclusion, the $\delta^{34}S$, TS and S^0 concentrations together provided a powerful set of indicators capable of detecting the invasion of sulfide in *Z. marina* and *P. oceanica*. Direct comparisons of $\delta^{34}S$ of plants from the different treatments were not possible due to considerable variation in $\delta^{34}S$ of the sulfur sources. The parameter F_{sulfide} adjusted for these differences and may therefore be an useful alternative, when $\delta^{34}S$ of the sulfur sources varies between study sites. In *Z. marina* the $\delta^{34}S$ and F_{sulfide} changed in all tissues in response to changes in sediment sulfide exposure, whereas *P. oceanica* only showed changes in roots and rhizomes. The TS concentrations increased in all tissues in both species, when exposed to increased sediment sulfide concentrations, however, only slightly in leaves of *P. oceanica*. Likewise concentrations of S^0 increased significantly in roots and rhizomes of both species at high sediment sulfide exposure. The indicators of sulfide invasion ($\delta^{34}S$, F_{sulfide} , TS and S^0) therefore generally showed a good correspondence. The two species differed in their response to increased sulfide exposure in that leaves of *P. oceanica* showed only weak indications of sulfide exposure compared to *Z. marina*. The two species were tolerant to sulfide exposure for several weeks but *P. oceanica* showed clear trends of reduced growth and reduced survival, after exposure to sulfide during 4 months. Future studies combining the indicators of sulfide invasion used in the present study with anatomical and physiological studies would provide important information on the sulfide dynamics in seagrasses.

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